Exact and Structural Methods for the Job Sequencing and Tool Switching Problem

Mid-internship report

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Abstract

The Job Sequencing and Tool Switching Problem (SSP) asks for an ordering of a set of jobs on a single flexible machine, together with a tool-loading policy for a capacitated tool magazine, that minimises the number of tool switches. This report develops *exact* and *structural* methods for the SSP. Two pillars organise the work, joined by a single geometric lens—the equivalence between the SSP and a generalised travelling-salesman problem over tool *configurations*. The first pillar is a structural analysis of the natural two-phase heuristic (group the jobs, then order the groups): we prove that the SSP optimum equals the minimum generalised-TSP cost over *all* feasible groupings, which shows the heuristic's optimality gap to be exactly the price of insisting on a minimum-cardinality grouping; we then give matching lower bounds, a family on which the gap grows without bound, an unconditional approximation-ratio bound, and two conjectures supported by exhaustive computation. The second pillar is exact computation: a branch-and-Benders-cut algorithm whose subproblem is the polynomial tool-loading oracle, and a family of position-indexed formulations with a fixed, polynomial row set together with an exact branch-and-price. We close with an objective assessment of why a configuration-based branch-and-price for the SSP is *bound-limited* rather than implementation-limited. Results established here are stated as theorems and propositions; results supported by exhaustive enumeration but not yet proved are stated as conjectures, with the scope of the supporting experiments made explicit. Longer proofs are collected in the appendix.

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1 Introduction

Motivation. A flexible manufacturing cell built around a single computer-numerically-controlled machine can produce a wide variety of parts because it draws on a large library of cutting tools. The machine, however, holds only a limited number of tools at once, in a magazine of capacity b; producing the full part mix therefore requires tools to be exchanged between jobs. Each exchange interrupts production while the magazine is reconfigured, so tool switches are the dominant controllable cost in such cells [10]. When the set of tools needed by each job is fixed, two decisions remain: in what *order* to process the jobs, and which tools to *keep or evict* at each step. The Job Sequencing and Tool Switching Problem (SSP) is the joint optimisation of these two decisions so as to minimise the total number of switches. The same structure—a capacitated store serving set-valued requests that may be reordered—recurs beyond machining, for example in printed-circuit-board assembly and in warehouse order picking, so the methods developed here are not specific to tools.

Why the problem is subtle. The two decisions are coupled in opposite directions. For a *fixed* job order, the optimal loading policy is known in closed form—the Keep Tool Needed Soonest (KTNS) rule of Tang and Denardo [31], computable in $O(n|T|)$ time—so the loading subproblem is easy. The difficulty lies entirely in the ordering: a good order places jobs with similar tool requirements next to one another so that few tools change hands, but "similarity" here is a set-overlap relation with no underlying metric, and the capacity constraint couples positions that are far apart in the sequence. The joint problem is NP-hard [10, 19], and even the strongest compact integer formulations [5, 11, 19] become difficult beyond moderate sizes. This structure—an easy inner problem nested inside a hard ordering problem—is precisely what decomposition methods are designed to exploit, and it motivates both pillars of this work.

Two pillars and a common lens. The report is organised around two complementary questions.

1. **How good is the natural heuristic?** A widely used two-phase method first partitions the jobs into the fewest groups whose tool requirements each fit in the magazine (the Job Grouping Problem, JGP), then orders the groups by solving a travelling-salesman-type problem, and finally reads off a schedule. We ask how far this heuristic can be from optimal, and when it is exact.
2. **How does one solve the SSP exactly, and how fast?** We develop a branch-and-Benders-cut algorithm and a family of position-indexed formulations with an exact branch-and-price, and assess their competitiveness.

Both pillars are most naturally expressed in one geometry. A *tool configuration* is a magazine-filling set of tools; the SSP is then a generalised travelling-salesman problem (GTSP) that must visit, in some order, one configuration able to serve each job, paying for the tools that change between consecutive configurations [7]. This configuration–GTSP equivalence (Section 2) is the thread tying the heuristic analysis to the exact formulations, and we establish it once, at the outset.

Contributions. The principal results are the following.

- *A reframing of the heuristic gap* (Section 3): we prove that the SSP optimum equals the minimum GTSP cost over *all* feasible groupings, not only minimum-cardinality ones. Consequently the two-phase heuristic's optimality gap is exactly the cost of restricting attention to minimum-cardinality groupings.
- *Gap quantification*: we prove the matching lower bounds $Z^* \geq \max(K^* - 1, |U| - b)$ and that the two-phase heuristic is a b-approximation; we exhibit a family on which the gap grows linearly without bound; and we state two conjectures—$H/Z^* \leq 4/3$ for $b = 3$, and gap $\leq K^* - 2$—each unfalsified by exhaustive enumeration over many thousands of instances.
- *A collapse theorem*: under a per-changeover magazine setup cost, the heuristic becomes exact once the ratio of setup cost to switch cost exceeds an explicit threshold.
- *Exact methods* (Section 4): a branch-and-Benders-cut solver whose subproblem is the polynomial KTNS oracle, verified optimal against exhaustive search; and position-indexed formulations (PCF, its symmetry-reduced variant PCF$'$, and the transition model PTF) with a fixed $O(n|T|)$ row set, their pricing subproblems, and an exact branch-and-price.
- *An objective competitiveness assessment* (Section 5): for the SSP the configuration relaxation collapses to the trivial coverage bound on most instances, so a configuration-based branch-and-price is bound-limited rather than implementation-limited.

Outline. Section 2 fixes notation, defines the SSP and KTNS, introduces the two-phase heuristic and the configuration–GTSP lens, proves the two structural lower bounds, fixes the switch-cost convention, and reviews the literature. Section 3 develops the structural theory of the heuristic gap; Section 4 presents the exact methods; Section 5 reports the experimental design, the verification of correctness, and the competitiveness assessment; and Section 6 concludes with open problems and future directions. Longer proofs are deferred to Appendix A.

2 Preliminaries

2.1 The Job Sequencing and Tool Switching Problem

We are given n jobs $J = \{1, \ldots, n\}$, m tools $T = \{1, \ldots, m\}$, a magazine capacity $b \in \mathbb{N}$ with $b < m$, and for each job j a set of required tools $T_j \subseteq T$ with $1 \leq |T_j| \leq b$. We write $U = \bigcup_{j \in J} T_j$ for the set of tools used by at least one job.

Definition 1 (Schedule and switch cost)**.** A *schedule* is a pair $(\pi, \mathbf{M})$ consisting of a permutation π of J and a sequence of *magazine states* $\mathbf{M} = (M_1, \dots, M_n)$, $M_i \subseteq T$, satisfying the capacity and feasibility constraints

$$|M_i| \le b \quad \text{and} \quad T_{\pi(i)} \subseteq M_i \qquad (i = 1, \dots, n).$$

With $M_0 = \varnothing$, its *switch cost* is the total number of tool insertions, $\sum_{i=1}^{n} |M_i \setminus M_{i-1}|$. The SSP asks for a schedule of minimum switch cost; this optimum is denoted Z^*.

Counting insertions is the standard cost model: under a fixed capacity each insertion into a full magazine forces exactly one removal, so insertions and removals differ only by boundary terms, which Section 2.6 makes precise.

Example 1 (A running instance: the 6-ring). We illustrate every definition on one instance, returned to throughout (Figure 1). The 6-*ring* has $n = m = 6$, capacity $b = 3$, and $T_j = \{j, j+1\}$ with indices taken modulo 6 (so $T_6 = \{6, 1\}$); the jobs form a cycle in which consecutive jobs share exactly one tool. Its job–tool incidence matrix (rows are tools 1–6, columns jobs 1–6; a blank denotes 0) is

	1	2	3	4	5	6
t_1	1					1
t_2	1	1				
t_3		1	1			
t_4			1	1		
t_5				1	1	
t_6					1	1

Process the jobs in the cyclic order $1, 2, 3, 4, 5, 6$ and load by the rule of Section 2.2. The magazine evolves as

$$\{1,2,3\} \;\to\; \{1,2,3\} \;\to\; \{1,3,4\} \;\to\; \{1,4,5\} \;\to\; \{1,5,6\} \;\to\; \{1,5,6\},$$

inserting 3 tools to build the first magazine and then one tool at each of positions $3, 4, 5$: a switch cost of 6, equivalently 3 once the initial fill is not charged (Section 2.6). Section 3 shows that 6 is optimal, yet the two-phase heuristic of Section 2.3 returns 7. The 6-ring is thus the smallest instance on which grouping-then-sequencing is provably sub-optimal, and it recurs below as the canonical witness of a positive gap.

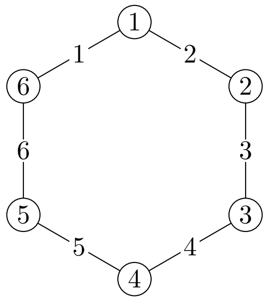

Figure 1: The 6-ring. Jobs 1–6 lie on a cycle; each edge is labelled by the single tool its two endpoints share ($T_j = \{j, j + 1\}$). Adjacent jobs overlap in one tool; non-adjacent jobs are tool-disjoint.

2.2 The tool-loading oracle

The inner problem—optimal loading for a fixed order—is solved by a greedy rule whose optimality is classical.

Proposition 1 (31)**.** *For a fixed permutation π, an optimal magazine sequence is produced by the* Keep Tool Needed Soonest *(KTNS) policy: whenever a tool must be inserted into a full magazine, evict a tool whose next required use lies furthest in the future (evicting any tool never required again). KTNS runs in $O(n|T|)$ time and returns the minimum switch cost over all loading policies for π.*

A proof appears in Tang and Denardo [31]; see also Crama et al. [10]. KTNS reduces the SSP to a pure sequencing problem, $Z^* = \min_\pi \mathrm{KTNS}(\pi)$, and is the exact polynomial subproblem exploited by the Benders algorithm of Section 4.2.

Remark 1 (The loading subproblem is offline optimal caching)*.* For a fixed order the loading problem is exactly offline paging: the magazine is a cache of size b, position i requests the set $T_{\pi(i)}$, and an insertion is a cache miss. Proposition 1 is then the tool-switching form of Bélády's optimal replacement rule—evict the item whose next request is furthest in the future—which minimises misses on any fixed request sequence [3]. The contrast with the full problem is instructive: paging fixes the request order, whereas the SSP may *reorder* the jobs, and it is precisely this freedom that turns a polynomially solvable caching problem into an NP-hard one. Every method in this report is, at heart, a way to search over reorderings while delegating the loading to this exact oracle.

2.3 Grouping, sequencing, and the two-phase heuristic

A complementary viewpoint ignores order and asks only which jobs may *share* a magazine.

Definition 2 (Job Grouping Problem)**.** A *group* is a set of jobs $G \subseteq J$ whose combined tool requirement fits in the magazine, $|\bigcup_{j \in G} T_j| \leq b$. The Job Grouping Problem (JGP) partitions J into the minimum number K^* of groups; equivalently, it covers all jobs by the fewest tool configurations of size at most b.

The JGP is itself NP-hard and is solved exactly by the asymmetric-representatives formulation of Jans and Desrosiers [18] or by the set-covering column generation of Crama and Oerlemans [9]. It induces the classical two-phase heuristic for the SSP:

1. solve the JGP to obtain K^* groups and a configuration serving each;
2. solve the *Group Sequencing Problem*—order the configurations by a travelling-salesman-type problem whose transition cost is the number of tools that change, a small instance handled by standard solvers [1, 17]—to obtain a sequence of configurations;
3. concatenate the jobs group by group and evaluate the resulting order with KTNS.

We write H for the switch cost this heuristic returns. Since it produces a feasible schedule, $H \geq Z^*$ always; the objects of study are the *gap* $H - Z^*$ and the *ratio* H/Z^* (Section 3).

Example 2 (Grouping the 6-ring)*.* On the running instance the JGP returns $K^* = 3$; one optimal grouping is $\{1, 2\}, \{3, 4\}, \{5, 6\}$, served by the configurations $\{1, 2, 3\}$, $\{3, 4, 5\}$, $\{5, 6, 1\}$. Any two of these three configurations differ in two tools, so every ordering of the groups changes two tools at each of its two transitions, and the best schedule the heuristic produces costs $H = 7$ (free-initial 4). This is one more than the optimum of 6 (free-initial 3) attained in Example 1: a gap of exactly one. The discrepancy is not a failure of the sequencing phase—the groups are ordered optimally—but of the *grouping* phase, which insists on only $K^* = 3$ groups; Section 3 makes this precise.

2.4 The configuration–GTSP lens

Both the heuristic and the exact formulations live in the space of configurations.

Definition 3 (Configurations and clusters)**.** A *configuration* is a set $C \subseteq T$ with $|C| = b$. Let $\mathcal{C} = \{C \subseteq T : |C| = b\}$, and for each job j let $H_j = \{C \in \mathcal{C} : T_j \subseteq C\}$ be the *cluster* of configurations able to serve j. For configurations C, C' put $d(C, C') = |C' \setminus C|$, the number of tools inserted in passing from C to C'.

Restricting to full magazines ($|C| = b$) loses no optimality: retaining a tool that may be useful later never increases cost. Under this view a schedule is a walk $C_{(1)}, \ldots, C_{(r)}$ in $\mathcal{C}$ that visits at least one configuration of every cluster H_j, with cost $\sum_l d(C_{(l)}, C_{(l+1)})$; the SSP is therefore a *generalised travelling-salesman problem with overlapping clusters* over $\mathcal{C}$ [7]. The clusters H_j overlap—a rich configuration serves many jobs—which is exactly what distinguishes the SSP from a textbook GTSP with disjoint clusters [13, 25, 27]. That a schedule and such a covering walk have equal cost is immediate in one direction—the distinct consecutive magazines of any schedule form a covering walk of the same cost—and the converse is established, in the stronger form ranging over *all* feasible groupings, as Theorem 1 of Section 3 (proof in Appendix A). This lens recurs throughout: it underlies the gap analysis of Section 3, the position-indexed formulations of Section 4.3, and the reading of the Benders master as a travelling-salesman skeleton (Section 4.2).

2.5 Two structural lower bounds

The following bounds are elementary but pervasive: they drive the gap theory of Section 3, calibrate the linear relaxations of Section 4, and serve as correctness checks for the computational study. Both are stated in the *free-initial* cost (Section 2.6), which does not charge the first magazine fill; the conversion to Definition 1 is the additive constant of Proposition 4.

Proposition 2 (Grouping bound)**.** *$Z^* \geq K^* - 1$.*

Proof. Fix any schedule and list its magazine states $M_1, \ldots, M_n$; collapse each maximal run of equal states to a single representative, obtaining a sequence $C_{(1)}, \ldots, C_{(r)}$ in which consecutive entries differ. Every job is served by some state, hence by some $C_{(l)}$, so $\{C_{(1)}, \ldots, C_{(r)}\}$ is a family of sets of size at most b covering all jobs. Assigning each job to one configuration that contains its requirement partitions J into groups, one per *distinct* configuration, each group having combined requirement contained in that configuration and hence of size at most b; this is a feasible grouping, so the number of distinct configurations is at least K^*. Therefore $r \geq K^*$. Finally, each of the $r - 1$ consecutive transitions changes at least one tool, so the free-initial switch cost is at least $r - 1 \geq K^* - 1$. □

Proposition 3 (Coverage bound)**.** *$Z^* \geq |U| - b$.*

Proof. Each tool of U is required by some job and therefore occupies the magazine at some position, so it is inserted at least once unless it belongs to the initial fill, which contains at most b tools. Hence at least $|U| - b$ insertions occur in the free-initial count. □

Corollary 1. *$Z^* \geq \max\{K^* - 1,\ |U| - b\}$.*

The two bounds are individually tight on some families and simultaneously slack on others (Section 3). The coverage bound $|U| - b$ is exactly the value to which the multicommodity-flow relaxation of da Silva and Yanasse [11] collapses—a fact central to the assessment of Section 5.4. The two are genuinely different measurements: on the 6-ring, $K^* - 1 = 2$ while $|U| - b = 6 - 3 = 3$, so the coverage bound is tight ($Z^* = 3$ free-initial, Example 1) and the grouping bound is slack, whereas on instances built from many nearly tool-disjoint groups the reverse holds. Neither dominates in general, and Section 3 shows that their maximum, though always valid, is not always attained.

2.6 Switch-cost convention

Formulations in the literature charge the initial magazine fill differently, and comparing their objective values naively produces spurious discrepancies; we therefore fix the convention.

Definition 4 (Empty-start and free-initial costs)**.** The *empty-start* cost charges every insertion from an initially empty magazine ($M_0 = \varnothing$, as in Definition 1). The *free-initial* cost does not charge the first fill; equivalently, it subtracts the $\min(b, |U|)$ tools loaded at the first position.

Proposition 4 (Convention identity)**.** *For every schedule,* $cost_{\text{empty}} = cost_{\text{free}} + \min(b, |U|)$.

Proof. An optimal loading (Proposition 1) keeps the magazine full, so the first position loads $\min(b, |U|)$ tools; the empty-start count charges these initial insertions while the free-initial count, by definition, omits them, and every later insertion is counted identically by both. □

The shift is a per-instance constant, so the gap $H - Z^*$ is convention-invariant whereas the ratio H/Z^* is not; ratio statements below are always taken in a single fixed convention. When objective values from different formulations are compared, they are first reduced to the empty-start convention via Proposition 4.

2.7 Related work

Exact formulations. Catanzaro et al. [5] give a family of compact integer formulations with a computational comparison, their strongest models adding overlap and linear-ordering inequalities. Laporte et al. [19] propose an integer program with a branch-and-cut. da Silva and Yanasse [11] give a multicommodity-flow formulation whose linear relaxation equals the coverage bound of Proposition 3. The configuration/GTSP formulation underlying our position-indexed models is developed by Colares and Wagler [7], Colares et al. [8].

Grouping and column generation. Crama and Oerlemans [9] introduce a set-covering formulation of the JGP solved by column generation with a strong relaxation; Jans and Desrosiers [18] give the asymmetric-representatives formulation. The pricing subproblem of the set-covering model is a Set-Union Knapsack Problem [16], NP-hard even under restrictive conditions. For branch-and-price we follow Barnhart et al. [2], Lübbecke and Desrosiers [20], Vanderbeck and Wolsey [32] and branch by the Ryan and Foster [30] same/different rule; subtour elimination uses the constraints of Miller et al. [22] with the lifting of Desrochers and Laporte [12], or lazily separated generalised cuts.

Adjacent exact studies. Two recent theses are close in method. Otiai [26] develops an exact branch-and-price for the Job Grouping Problem with Ryan–Foster branching and finds the set-covering relaxation strong enough to solve benchmark instances in seconds where the compact asymmetric-representatives model times out. Moreira [24] studies the SSP through a magazine-configuration formulation in the sense of Colares and Wagler [7] together with a configuration-generation heuristic, and measures the strength of that formulation's linear relaxation against the compact models; we revisit those measurements in Section 5.4.

Heuristics. Constructive and metaheuristic upper bounds include the branch-and-bound and bounding ideas of Ghiani et al. [15] and the hybrid genetic search of Mecler et al. [21], which seeds its population from JGP solutions.

3 Structural analysis of the heuristic gap

The two-phase heuristic was seen on the running instance to cost one more than the optimum (Example 2), with the loss occurring in the grouping phase. This section makes that observation precise and general. We first prove that admitting *arbitrary* groupings renders the configuration framework exact, so the heuristic's entire gap is the price of insisting on a minimum-cardinality grouping (Section 3.1); we then show the gap is genuinely present and can grow without bound (Section 3.2), prove that the heuristic is nonetheless a b-approximation and state two sharpening conjectures (Section 3.3), exhibit a cost regime in which the gap vanishes (Section 3.4), and isolate the algorithmic question this leaves open (Section 3.5).

3.1 The gap is the price of minimum-cardinality grouping

We attach a cost to a grouping by ordering its configurations optimally.

Definition 5 (Feasible grouping and its cost)**.** A *feasible grouping* is a partition $\mathcal{P} = \{G_1, \ldots, G_p\}$ of J together with, for each group, a configuration $C_l \supseteq \bigcup_{j \in G_l} T_j$ (which exists iff $|\bigcup_{j \in G_l} T_j| \leq b$). Its *cost* is the cheapest free-initial walk through its configurations,

$$\gamma(\mathcal{P}) = \min_{\sigma \in S_p} \sum_{l=1}^{p-1} d\big(C_{\sigma(l)}, C_{\sigma(l+1)}\big).$$

The Job Grouping Problem seeks a feasible grouping of minimum cardinality K^*, and the two-phase heuristic evaluates γ on such a grouping. Removing the cardinality restriction makes the framework exact.

Theorem 1 (Grouping exactness)**.** *$Z^* = \min_{\mathcal{P}} \gamma(\mathcal{P})$, the minimum taken over* all *feasible groupings, of every cardinality.*

The proof (Appendix A) is a pair of constructions: any feasible grouping, processed group by group, is a schedule of cost $\gamma(\mathcal{P})$, giving "$\leq$"; and any optimal schedule, its jobs grouped by the magazine under which they run, is a feasible grouping of no greater cost, giving "$\geq$". The identity was confirmed by exhaustive enumeration of 1,306 instances without exception.

Corollary 2 (The heuristic gap)**.** *Writing $H = \min\{\gamma(\mathcal{P}) : |\mathcal{P}| = K^*\}$ for the best two-phase cost,*

$$H - Z^* = \min_{|\mathcal{P}|=K^*} \gamma(\mathcal{P}) \; - \; \min_{\mathcal{P}} \gamma(\mathcal{P}) \; \geq \; 0,$$

and the gap is strictly positive exactly when no minimum-cardinality grouping attains the overall minimum of γ.

Remark 2 (A zero-gap certificate). Since $H \geq Z^* \geq \max(K^* - 1, |U| - b)$ (Corollary 1), any instance on which the heuristic attains a lower bound, $H = \max(K^* - 1, |U| - b)$, has zero gap. This already disposes of the k-rings with $k \in \{3, 4, 5\}$ (Table 1), where $H = |U| - b$.

Example 3 (Where the ring loses). The optimum of the 6-ring is realised by the *four*-configuration grouping $\{1, 2, 3\} \to \{1, 3, 4\} \to \{1, 4, 5\} \to \{1, 5, 6\}$, consecutive configurations differing in a single tool, for $\gamma = 3$ (cf. Example 1). Every minimum-cardinality grouping uses only $K^* = 3$ configurations and costs 4 (Example 2). The gap of one is precisely the difference in Corollary 2: the cheaper grouping is not of minimum cardinality, so the grouping phase never offers it to the sequencer.

k	K^*	Z^*	H	gap	H/Z^*
3	1	0	0	0	—
4	2	1	1	0	1
5	3	2	2	0	1
6	3	3	4	1	4/3
7	4	4	5	1	5/4
8	4	5	6	1	6/5

Table 1: The k-ring ($b = 3$, free-initial costs).

3.2 The gap is real and unbounded

Table 1 collects the k-ring ($T_j = \{j, j+1\}$ modulo k, $b = 3$) for $k = 3, \ldots, 8$: the gap first appears at $k = 6$, where the ratio is largest. Replicating the worst ring makes the gap arbitrarily large.

Proposition 5 (Unbounded gap)**.** *Let R_g be g tool-disjoint copies of the 6-ring ($b = 3$, $|U| = 6g$). Then $Z^* = 6g - 3$ and $H = 7g - 3$, so the gap $H - Z^* = g$ is unbounded while the ratio $(7g - 3)/(6g - 3)$ decreases from $4/3$ at $g = 1$ towards $7/6$.*

The proof (Appendix A) pairs the coverage bound $Z^* \geq |U| - b = 6g - 3$ with a sequential schedule attaining it, and counts the heuristic's cost as g within-copy contributions of 4 and $g - 1$ inter-copy transitions of 3. The values were verified for $g = 1, \ldots, 4$.

Remark 3 (Perfectness does not predict the gap)*.* One might hope that polyhedral regularity of an instance governs its gap; it does not. Form the *conflict graph* on the jobs, joining i and j when they cannot share a configuration ($|T_i \cup T_j| > b$). For the 6-ring this graph is the triangular prism, the complement of C_6, which is perfect, yet the gap is one; for the 5-ring it is the odd cycle C_5, which is imperfect [6], yet the gap is zero. Perfectness of the conflict graph and the size of the gap are therefore independent.

3.3 A worst-case guarantee and two conjectures

Although unbounded in absolute terms, the gap is bounded in ratio.

Proposition 6. *$H \leq b(K^* - 1)$.*

Proof. A minimum-cardinality grouping has K^* configurations; any ordering of them has $K^* - 1$ transitions, each changing at most b tools since every configuration has size b. Hence $\gamma \leq b(K^* - 1)$ for that grouping, and H is no larger. □

Theorem 2 (b-approximation)**.** $\dfrac{H}{Z^*} \leq \dfrac{b(K^* - 1)}{\max(K^* - 1,\ |U| - b)} \leq b.$

Proof. Immediate from Proposition 6 and Corollary 1, since $\max(K^* - 1, |U| - b) \geq K^* - 1$. □

The true worst case appears far smaller. Exhaustive enumeration supports two sharper statements.

Conjecture 1 ($b = 3$ ratio)**.** For every instance with $b = 3$, $H/Z^* \leq 4/3$.

Checked by exhaustive enumeration of the $b = 3$ two-tools-per-job family with $m \leq 6$ (10,691 instances) and of mixed size-$\{2, 3\}$ jobs with $m \leq 5$ (a further 6,065): the maximum ratio is exactly $4/3$, attained only by the 6-ring, with no counterexample.

Conjecture 2 (gap bound)**.** For every instance with $K^* \geq 2$, $H - Z^* \leq K^* - 2$.

No violation arose over ∼1,380 enumerated instances; it is consistent with $K^* \leq 2$ forcing zero gap, and with the family of Proposition 5, where $K^* = 3g$ and the gap g satisfies $g \leq 3g - 2$.

Remark 4 (Dependence on b). The constant 4/3 is specific to $b = 3$. The extremal ratio is non-decreasing in b, reaching 3/2 at $b = 5$ (an instance with $K^* = 4$, $|U| = 9$, $Z^* = 4$, $H = 6$); a tight general-b bound is open.

3.4 Collapse under a setup cost

In practice each changeover incurs a fixed setup time beyond the per-tool exchanges. Model this by charging, in addition to the tool switches, a cost $\rho \geq 0$ per *configuration change*; the objective becomes switches $+ \rho \cdot$ (#configuration changes), where ρ is the ratio of setup time to unit switch time. A grouping into p configurations makes $p-1$ changes, so the setup term alone is minimised by the Job Grouping Problem; as ρ grows it dominates and pulls the optimum onto minimum-cardinality groupings.

Theorem 3 (Setup-cost collapse)**.** *If $\rho > H - Z^*$, then every optimal solution of the setup-augmented objective uses a minimum-cardinality grouping, and the two-phase heuristic is exact. Hence the collapse threshold satisfies $\rho_c \leq H - Z^*$; for the ring family $\rho_c = 1$.*

Proof. Every feasible grouping $\mathcal{P}$ has $|\mathcal{P}| \geq K^*$ and, by Theorem 1, $\gamma(\mathcal{P}) \geq Z^*$, so its augmented cost is $\rho(|\mathcal{P}| - 1) + \gamma(\mathcal{P}) \geq \rho(|\mathcal{P}| - 1) + Z^*$. The best minimum-cardinality grouping has augmented cost $\rho(K^*-1)+H$. If $|\mathcal{P}| > K^*$, the former exceeds the latter whenever $\rho(|\mathcal{P}|-K^*) > H - Z^*$, which holds for all such $\mathcal{P}$ as soon as $\rho > H - Z^*$ (since $|\mathcal{P}| - K^* \geq 1$). Thus no over-cardinality grouping is optimal, and the heuristic—which selects the cheapest ordering of a minimum-cardinality grouping—attains the optimum. The ring value $\rho_c = 1$ follows from $H - Z^* = 1$ and the exact ring costs. □

Corollary 3. *On ring-like instances the setup-to-switch ratios $\rho \approx$ 3–60 reported for real tooling [28, 29] far exceed ρ_c, so the two-phase heuristic is exactly optimal for the setup-augmented objective there.*

3.5 Which grouping to choose

Theorem 1 converts the heuristic's weakness into a concrete algorithmic question. Since the optimum is attained by *some* feasible grouping, and minimum-cardinality ones may miss it (Example 3), one should ask on *which* grouping to run the sequencing phase. Admitting a single extra group already recovers the ring optimum; in general the trade-off is between more configuration changes and cheaper individual transitions. Characterising the groupings that attain $\min_{\mathcal{P}} \gamma(\mathcal{P})$, or giving an efficient rule that improves on the minimum-cardinality choice, is open and is the most direct route to closing the gap in practice.

4 Exact methods

Exact methods for the SSP follow two complementary routes. The first keeps the sequencing variables and *decomposes*: a master problem fixes the order while a subproblem prices the loading. Because the loading subproblem is polynomial (Proposition 1), this decomposition can be driven to a certified optimum with exact cuts (Section 4.2). The second *reformulates* the problem over configurations in a form whose row set is fixed and whose columns are generated (Section 4.3). We first record the compact baselines against which both are measured.

4.1 Compact formulations and the coverage relaxation

Several compact integer formulations are available. Catanzaro et al. [5] give a family F1–F5 in job–position and ordering variables, the tighter members adding linear-ordering and overlap inequalities; we use F4 as a baseline. Laporte et al. [19] formulate the problem with tool-state variables and a branch-and-cut. da Silva and Yanasse [11] give a multicommodity-flow model in which each tool routes one unit of flow through the position network. The last is singled out by its relaxation.

Proposition 7. *The linear relaxation of the multicommodity-flow formulation equals the coverage bound $|U| - b$ of Proposition 3.*

Established by da Silva and Yanasse [11] and reconfirmed here, this matters twice: the compact relaxation is no stronger than the elementary counting bound; and since $|U| - b$ equals Z^* on many families (Section 3), the model is fast exactly where that bound is tight and weak where it is not—a dichotomy quantified in Section 5.4.

4.2 A branch-and-Benders-cut algorithm

Idea. A schedule is an order together with a loading. Once the order is fixed, the minimum loading cost is computed exactly and in polynomial time by KTNS; so we place the *order* in an integer master and represent the loading cost by a single continuous variable θ, refined by cuts generated from the exactly solved loading subproblem. In contrast to column generation, whose pricing subproblem for this problem is NP-hard (Section 2.7), the Benders subproblem here is polynomial, so every cut is exact.

Master. Introduce a depot 0 and arc variables $x_{ij} \in \{0, 1\}$ for $i, j \in \{0, 1, \dots, n\}$, $i \neq j$, with $x_{ij} = 1$ when job j is processed immediately after i (the depot marking start and end). The assignment constraints

$$\sum_{i \neq j} x_{ij} = 1 \quad (\forall j), \qquad \sum_{j \neq i} x_{ij} = 1 \quad (\forall i),$$

together with subtour-elimination constraints $\sum_{i,j \in S} x_{ij} \leq |S| - 1$ for $S \subseteq \{1, \dots, n\}$, make the x a Hamiltonian order of the jobs. The master is

$$\min\ \theta \qquad \text{s.t.} \qquad \theta \geq \textstyle\sum_{i,j} w_{ij}\, x_{ij}, \quad \text{assignment, subtour, and Benders cuts,}$$

in which θ lower-bounds the loading cost and the subtour and Benders constraints are separated lazily.

An initial valid inequality. At the boundary between consecutive jobs i and j at least $w_{ij} := \max\big(0, |T_i \cup T_j| - b\big)$ tools are switched: the magazine holds at most b tools yet must serve first T_i and then T_j, so the members of $T_i \cup T_j$ absent at position i—at least $|T_i \cup T_j| - b$ of them—are inserted by position j. Summing over the chosen arcs gives the valid inequality $\theta \geq \sum_{ij} w_{ij} x_{ij}$, used to initialise the master.

Subproblem. Fix the order and relabel the jobs $1, \dots, n$ along it. With $g_{t,i} \in [0, 1]$ indicating that tool t occupies the magazine at position i and $s_{t,i} \geq 0$ its insertion there, the loading cost is the optimal value of

$$\begin{aligned}
\min\ & \sum_{t,i} s_{t,i} \\
\text{s.t. } & g_{t,i} = 1 && (t \in T_i), \\
& \textstyle\sum_t g_{t,i} \leq b && (\forall i), \\
& s_{t,i} \geq g_{t,i} - g_{t,i-1}, \quad s_{t,i} \geq 0 && (\forall t, i),
\end{aligned}$$

with $g_{t,0} = 0$. The constraint matrix has the consecutive-ones property and is totally unimodular, so the relaxation is integral and, by Proposition 1, its optimum equals KTNS of the order. Writing R_i for the tool set of the job at position i, an optimal dual prices each requirement "tool t present at position i" by a multiplier $\bar{\alpha}_{t,i}$ and each capacity bound by $\bar{\beta}_i \geq 0$; linear-programming duality then yields the Benders optimality cut

$$\theta \;\geq\; \sum_i \sum_{t \in R_i} \bar{\alpha}_{t,i} \;-\; b \sum_i \bar{\beta}_i$$

(the multipliers of the unit upper bounds on the $g_{t,i}$ contribute a further term, suppressed here). The requirement multipliers carry the dependence on the order—through which job, hence which R_i, occupies each position—so the right-hand side is linear in the master's assignment variables; and because the dual feasible region does not depend on the order, the cut is globally valid, the defining property of a Benders cut.

Cut families. Two kinds of cut are separated, and the study ablates them. *(i) Benders optimality cuts* from the dual above, separated at integer master solutions and, optionally, at fractional ones (*fractional cuts*) to tighten the bound before branching. *(ii) Combinatorial KTNS cuts*: at an integer order $\bar{\sigma}$ with $\text{KTNS}(\bar{\sigma}) = \bar{z}$, the inequality

$$\theta \;\geq\; \bar{z}\Big(1 - \sum_{(i,j) \in A(\bar{\sigma})} (1 - x_{ij})\Big)$$

is valid—it forces $\theta \geq \bar{z}$ when the order is exactly $\bar{\sigma}$ (all arcs $A(\bar{\sigma})$ selected) and is vacuous otherwise—and uses only the polynomial oracle, with no dual computation. Finally, *triplet bounds* generalise w_{ij} to ordered triples of jobs, contributing $O(n^3)$ valid inequalities that strengthen the root relaxation.

Correctness. The subproblem's value equals KTNS by total unimodularity and Proposition 1; this was confirmed numerically (the loading program and the combinatorial oracle agreeing on all 60 random tests), and the resulting solver returns the exhaustive-search optimum on every small instance. The cut families form a full-factorial design—combinatorial versus dual cuts, fractional separation on or off, triplet bounds on or off—whose ablation is reported in Section 5.2.

4.3 Position-indexed formulations

Motivation. The configuration view of Section 2.4 invites a column-generation model whose columns are configurations. A naive master indexes configurations by sequence position and eliminates subtours with Miller–Tucker–Zemlin constraints; but those constraints reference the configuration variables, so the row set grows and changes as columns are added. We seek instead masters whose row set is *fixed* and polynomial in $(n, |T|)$, so that column generation only ever adds columns to existing rows. Two such formulations are developed.

Position–configuration formulation (PCF). Let $y_C^k \in \{0,1\}$ indicate that configuration C occupies position k ($k = 1, \dots, n$) and let $w_t^k \geq 0$ count the insertion of tool t at position k:

$$\begin{aligned}
\min \quad & \sum_{t,k} w_t^k \\
\text{(P)} \quad & \sum_{C} y_C^k = 1, \quad k = 1, \dots, n; \\
\text{(Cov)} \quad & \sum_{k} \sum_{C \supseteq T_j} y_C^k \geq 1, \quad j \in J; \\
\text{(W)} \quad & w_t^k \geq \sum_{C \ni t} (y_C^k - y_C^{k-1}), \quad t \in T,\ k \geq 2; \\
\text{(T)} \quad & \sum_{k \geq 2} w_t^k + \sum_{C \ni t} y_C^1 \geq 1, \quad t \in U.
\end{aligned}$$

The configuration variables y are generated as columns; the rows (P), (Cov), (W), (T) number $O(n|T|)$ and never change. Constraint (P) places one configuration per position, (Cov) serves every job, and (W) charges an insertion of t whenever it enters the magazine.

Proposition 8. *PCF is exact: its integer feasible points are exactly the SSP schedules, of equal cost. Its linear relaxation* without *(T) equals* 0*;* with *the counting rows (T) it equals* $|U| - b$*.*

The vanishing of the plain relaxation is the fractional pathology of Tang and Denardo [31]: a single convex blend of configurations, repeated at every position, satisfies (P), (Cov) and (W) at zero cost. The counting rows (T)—each tool of U is inserted at least once unless present at position one—restore the coverage bound, matching the multicommodity model (Proposition 7) with one extra family of rows; both facts were confirmed computationally on the ring and on random instances.

Position–transition formulation (PTF). To exceed the coverage bound we index *transitions* rather than positions. Augment the configurations with an absorbing tool-less node $\perp$, used to pad schedules that change configuration fewer than $n-1$ times, and let $z_{C,C'}^k \geq 0$ be a column for the step $k \to k+1$ that leaves configuration C and enters C'. The formulation is

$$\begin{aligned}
\min \quad & \sum_{k=1}^{n-1} \sum_{(C,C')} d(C,C')\, z_{C,C'}^k \\
\text{(S)} \quad & \sum_{(C,C')} z_{C,C'}^k = 1, \qquad k = 1, \dots, n-1; \\
\text{(F)} \quad & \sum_{(C,C')\colon t \in C'} z_{C,C'}^k = \sum_{(D,D')\colon t \in D} z_{D,D'}^{k+1}, \qquad t \in T,\ k = 1, \dots, n-2; \\
\text{(Cov)} \quad & \sum_{k=1}^{n-1} \sum_{(C,C')\colon T_j \subseteq C} z_{C,C'}^k + \sum_{(C,C')\colon T_j \subseteq C'} z_{C,C'}^{n-1} \ \geq\ 1, \qquad j \in J.
\end{aligned}$$

Constraint (S) selects one transition per step; the flow-consistency rows (F) require, tool by tool, that what enters position $k+1$ at the end of step k is what leaves it at the start of step $k+1$, so the columns describe a single coherent walk $C^1 \to \cdots \to C^n$; and (Cov) serves every job at the configuration of some position—the tail of a step, or the head of the last. The objective charges $d(C,C') = |C' \setminus C|$ per transition. As in PCF the rows number $O(n|T|)$ and are fixed, while the transition columns are generated.

Proposition 9. *PTF is exact, and its linear relaxation is at least* $|U| - b$ *and can be strictly greater.*

On an instance whose optimum exceeds both bounds of Corollary 1, the PTF relaxation returns 2.10 against a coverage bound of 2; on the 6-ring it is tight at $Z^* = 3$ (both computed exactly). PTF is thus the first relaxation in this family to improve on the elementary bound, and it does so with a fixed, polynomial row set.

Symmetry reduction: the formulation PCF′. PCF carries a large *padding* symmetry. A schedule using $m < n$ distinct configurations is encoded by repeating configurations to fill all n positions, and the repeats may sit anywhere, so one physical schedule has many equal-cost encodings—distinct nodes to a tree search. We remove this freedom by relaxing the assignment equation (P) to an inequality and forcing the empty positions to a suffix:

$$(\mathrm{P}') \quad \sum_C y_C^k \le 1 \ \ (\forall k); \qquad (\mathrm{G}) \quad \sum_C y_C^{k+1} \le \sum_C y_C^k \ \ (k = 1, \dots, n-1);$$

with (Cov), (W), (T) unchanged. Call this formulation PCF′. At an integer point the counts $\sum_C y_C^k$ are non-increasing in k, so the *used* positions form a prefix $1, \dots, p$ and the empties a suffix, with the cost-free first configuration at position 1. The contiguity rows (G) are not decorative: without them an interior empty position would make (W) read the re-entry as a full reload and (T) misattribute the free load, so (G) is what keeps the relaxed model faithful.

Proposition 10. *PCF′ is exact, and its linear relaxation equals that of PCF—$|U| - b$ with the counting rows (T), and* 0 *without. It removes symmetric integer encodings but does not strengthen the bound.*

The proof is in Appendix A. The separation of concerns is deliberate: PCF′ attacks symmetry, and only PTF attacks the bound.

Column generation and robust branching. Both PCF′ and PTF have $O(n|T|)$ rows but exponentially many columns ($\binom{|T|}{b}$ configurations, or arcs), so their relaxations are solved by *column generation*: a restricted master over a small column pool is reoptimised, and a pricing subproblem repeatedly supplies a column of most-negative reduced cost until none remains. Driving this at every node of a branch-and-bound tree gives *branch-and-price* [2, 20]. Branching on a single column y_C^k is ineffective under the position symmetry and forces the pricer to carry a growing list of forbidden columns. We branch instead on the *presence* variables $a_t^k = \sum_{C \ni t} y_C^k \in \{0, 1\}$ already appearing in (W): fixing a_t^k to 0 or 1 forbids or forces tool t at position k, which in the pricing subproblem merely shifts one dual weight and leaves the oracle's *form* unchanged. This *robust* branching keeps pricing and branching compatible at every node, and the resulting algorithm returns the optimum Z^* on the ring and on random instances.

Pricing PCF′: a b-subset with coverage bonuses. Let $\alpha_k, \gamma_k \le 0$, $\pi_j, \mu_{t,k}, \lambda_t \ge 0$ be the duals of (P′), (G), (Cov), (W), (T). Direct dual accounting (collecting the coefficient of y_C^k, which enters (W) and (T) through $a_t^k = \sum_{C \ni t} y_C^k$) gives the reduced cost

$$\bar{c}(C, k) = \beta_k - \sum_{t \in C} \rho_t^k - \sum_{j:\, T_j \subseteq C} \pi_j, \qquad \rho_t^k = -\mu_{t,k}[k \ge 2] + \mu_{t,k+1}[k \le n-1] + \lambda_t\,[k = 1,\ t \in U],$$

with β_k depending only on the position. For each k the most-negative reduced cost is obtained by maximising the configuration-dependent part,

$$\max_{|C| = b} \sum_{t \in C} \rho_t^k + \sum_{j:\, T_j \subseteq C} \pi_j,$$

i.e. choosing b tools to collect per-tool rewards plus a bonus for each job whose *entire* requirement lands inside C. This is a Set-Union Knapsack Problem [16]—the grouping-pricing class of Crama and Oerlemans [9]—solved by enumerating the $\binom{|T|}{b}$ subsets at magazine sizes, or by a compact mixed-integer program. The branching dual of a_t^k enters additively in ρ_t^k, so branching does not change the oracle. This reduced cost was verified against the solver's own dual values on the ring and several random instances (matching to machine precision)—a process that caught and corrected a sign error in ρ_t^k in an earlier derivation.

Pricing PTF: a coupled pair of subsets. For PTF a column is an arc (C, C'), and its reduced cost contains the bilinear term $\sum_t [t \in C' \setminus C]\,(1-\lambda_t)$ together with chaining-dual terms on the tools of C and C'. The pricer cannot therefore separate into a single set: it must choose *two* b-subsets $C \neq C'$ at once, a bilinear selection linearised by McCormick on the products $x'_t(1-x_t)$ and solved as a mixed-integer program with $2|T|$ binaries. This heavier pricing is the algorithmic price of PTF's stronger bound, and its reduced cost was likewise verified against the solver duals. The contrast—PCF′ pairing a weak bound with a cheap single-set pricer, PTF a stronger bound with a costly coupled-pair pricer—is the central trade-off assessed in Section 5.4.

4.4 Configuration-level subtour constraints

A configuration-indexed travelling-salesman model with Miller–Tucker–Zemlin (MTZ) subtour constraints [12, 22] admits two natural levels of aggregation, which differ in correctness.

Proposition 11. *Per-configuration MTZ constraints (one potential per configuration) are exact. Cluster-aggregated MTZ constraints (one potential per job) are not: they can exclude the optimum.*

The aggregated form fails under job domination: if $T_i \subseteq T_j$, every configuration serving j also serves i, so a transition between two such configurations places $\{i, j\}$ at both ends and forces the potentials of i and j into a two-cycle that no ordering can satisfy. A four-job instance with $b = 3$ has an optimum of 1 that the aggregated model values at 2. One should therefore use the per-configuration form or lazily separated generalised subtour cuts; the position-indexed formulations above sidestep the issue, carrying no subtour constraints at all.

5 Computational study

The methods of Section 4 have been implemented and are being benchmarked. We report the correctness verification, which is complete, and the experimental design; the large-scale comparison is in progress, and its tables are the report's only deferred content.

5.1 Verification of correctness

Because the structural results and the solvers all rest on the loading cost, the cost evaluator is validated first. KTNS (Proposition 1) was checked against an independent exact dynamic program over the magazine state on 8,000 random sequences, with no disagreement. Each exact method—the branch-and-Benders-cut solver, the compact formulations of Section 4.1, and the position-indexed branch-and-price—was then required to return the exhaustive-search optimum on small instances; this cross-solver regression net exposes a formulation error as a disagreement on a tiny instance, and is run before any campaign. For the position-indexed models in particular, exactness and the relaxation values of Propositions 8 and 10 were checked by full configuration enumeration on the ring, and each pricing reduced cost was matched to the solver's own dual values before the branch-and-price was trusted. All objective values are reduced to the empty-start convention through Proposition 4 before comparison, so that models with different native conventions are placed on one scale.

5.2 Experimental design

The benchmark draws on the standard SSP instance families of Catanzaro et al. [5] ($n = 8$–15 jobs, $m = 5$–15 tools, $b = 3$–5), Crama et al. [10] ($n = 10$, $m = 10$, $b = 4$) and Laporte et al. [19] ($n = 8$–15, $m = 15$, $b = 5$), spanning a range of tool densities and capacities. Three things are measured. *(i)* The branch-and-Benders-cut solver is run as a full-factorial ablation of its cut families—combinatorial against dual optimality cuts, fractional separation on or off, triplet bounds on or off—against the three compact baselines. *(ii)* The position-indexed branch-and-price is run in its PCF and PTF variants. *(iii)* For every solver and instance the *root linear-relaxation value* is recorded alongside the optimum, so that bound tightness can be studied directly. The common metric is the empty-start KTNS cost of the returned sequence, well defined for every method regardless of its native objective. Runs are single-threaded for reproducible timing, with a per-instance time limit, and are sharded across a compute cluster.

5.3 Results

The campaigns are executing at the time of writing. The completed study will report, across the instance families: solve rates and running times for each method; the marginal contribution of each cut family in the branch-and-Benders-cut ablation; the comparison of the position-indexed branch-and-price against the compact baselines on the shared metric; and the distribution of the root-relaxation gap, which the following subsection argues is the decisive quantity.

5.4 The competitiveness of configuration branch-and-price

A guiding question of the exact-methods pillar is whether a configuration-based branch-and-price can be made fast for the SSP, and the structural results already constrain the answer. By Propositions 7 and 8 the configuration relaxation collapses to the coverage bound $|U| - b$, and by the gap theory of Section 3 this bound equals Z^* on a large part of the instance space. Where the root bound already equals the optimum branching is unnecessary; where it does not, a relaxation no stronger than $|U| - b$ forces a search tree that faster pricing or branching cannot avoid. The contrast with the Job Grouping Problem is sharp: there the set-covering relaxation is strong enough that an analogous branch-and-price closes benchmark instances in a handful of nodes [26], whereas for the SSP the corresponding configuration relaxation does not improve on the trivial bound on most instances [24]. A configuration branch-and-price for the SSP is therefore *bound-limited* rather than implementation-limited. The position–transition formulation, the first in the family to exceed $|U| - b$ (Proposition 9), is exactly the structural lever this diagnosis points to; but it pays off only where the gap opens, and broadening that improvement is taken up in Section 6.

6 Conclusions and future directions

6.1 Summary

A single device—the equivalence between the SSP and a generalised travelling-salesman problem over tool configurations—has organised both a structural theory of the standard heuristic and a family of exact methods. On the structural side we proved that admitting arbitrary groupings makes the configuration framework exact (Theorem 1), so that the two-phase heuristic's gap is exactly the price of a minimum-cardinality grouping (Corollary 2); we bounded that gap from below and above (Corollary 1, Theorem 2), showed it can grow without bound (Proposition 5), and exhibited a setup-cost regime in which it vanishes identically (Theorem 3). Two conjectures—a $4/3$ ratio at $b = 3$ and a gap of at most $K^* - 2$—survived exhaustive testing. On the algorithmic side we gave a branch-and-Benders-cut solver whose subproblem is exact and polynomial, and position-indexed formulations with a fixed row set, among which the position–transition model is the first configuration relaxation to improve on the coverage bound. The methods are implemented and verified against exhaustive search; their large-scale comparison is under way.

6.2 Future directions

Closing the gap theory. The most consequential open question is the gap bound $H - Z^* \leq K^* - 2$ (Conjecture 2): a proof, a proof of the $K^* = 3$ case for all b, or a counterexample would each settle the worst-case behaviour of the standard heuristic, which the literature has not analysed. Its constructive counterpart is the grouping-selection problem of Section 3.5—since some grouping attains the optimum, can one be found, or approximated, more cheaply than by solving the SSP outright?

Stronger relaxations. The competitiveness analysis (Section 5.4) locates the obstacle to a fast exact configuration method in the weakness of the $|U| - b$ relaxation. The position–transition formulation breaks that barrier on some instances (Proposition 9); a systematic study of its valid inequalities and facets, and of the instances on which its bound exceeds $|U| - b$, is the most direct route to an exact method competitive in time and not only in correctness.

The position–transition device beyond the SSP. Nothing in the position–transition construction is specific to tool switching except the transition metric: position-transition columns with per-element consistency rows apply to any generalised travelling-salesman problem with overlapping clusters [27]. Whether the relaxation strength observed here persists in that generality is open, and an affirmative answer would carry the contribution beyond the SSP.

Approximation and online variants. The ratio theory of Section 3 should be reconciled with worst-case approximation guarantees reported elsewhere for the SSP, read critically as to the bound and convention against which they hold. Separately, the caching equivalence of Remark 1 suggests an online SSP, in which jobs arrive over time, analysed competitively against paging machinery—with the offline loading oracle as a natural baseline.

Learning-augmented column generation. Machine-learning acceleration of pricing and branching [4, 14, 23] is an active but crowded field. It is worth pursuing here only if the benchmark data expose a concrete learning signal—such as the pronounced imbalance between zero-gap and positive-gap instances that the structural results predict—rather than as an end in itself.

A Proofs

Proof of Theorem 1. Both inequalities are explicit constructions; by Proposition 1 we may assume the optimal schedule loads by KTNS, so every magazine is full, $|M_i| = b$.

($\leq$). Let $\mathcal{P} = \{G_1, \dots, G_p\}$ be a feasible grouping with configurations $C_1, \dots, C_p$, and let σ attain the minimum in Definition 5. Process the groups in the order $G_{\sigma(1)}, \dots, G_{\sigma(p)}$, and within each group its jobs in any order, holding magazine $C_{\sigma(l)}$ throughout group $G_{\sigma(l)}$. This is feasible, since each $j \in G_{\sigma(l)}$ satisfies $T_j \subseteq \bigcup_{i \in G_{\sigma(l)}} T_i \subseteq C_{\sigma(l)}$ and $|C_{\sigma(l)}| = b$. The magazine is constant within a group and changes only at the $p - 1$ group boundaries, so the free-initial switch cost is $\sum_{l=1}^{p-1} d(C_{\sigma(l)}, C_{\sigma(l+1)}) = \gamma(\mathcal{P})$. Hence $Z^* \leq \gamma(\mathcal{P})$, and minimising over $\mathcal{P}$ gives $Z^* \leq \min_{\mathcal{P}} \gamma(\mathcal{P})$.

($\geq$). Let $(\pi, \mathbf{M})$ be an optimal schedule with full magazines. Partition J into the maximal blocks of consecutive positions sharing one magazine; let these be $B_1, \dots, B_r$ in order, with (size-b) magazines $D_1, \ldots, D_r$ and $D_l \neq D_{l+1}$. Each block is a group whose jobs run under D_l, so $\bigcup_{j \in B_l} T_j \subseteq D_l$ with $|D_l| = b$; the blocks thus form a feasible grouping $\mathcal{P}'$ with configurations $D_1, \ldots, D_r$. The identity ordering of these configurations is one admissible σ, so $\gamma(\mathcal{P}') \leq \sum_{l=1}^{r-1} d(D_l, D_{l+1})$, which is exactly the free-initial switch cost of $(\pi, \mathbf{M})$, namely Z^*. Therefore $\min_{\mathcal{P}} \gamma(\mathcal{P}) \leq \gamma(\mathcal{P}') \leq Z^*$. Combining the two inequalities gives equality. □

Proof of Proposition 5. The copies use pairwise disjoint tool sets, so $|U| = 6g$ and Proposition 3 gives $Z^* \geq 6g - 3$. For a matching schedule, process the copies one after another, each in the cyclic order of Example 1. Within a copy that loading makes 3 insertions to build the first magazine and 3 more across the copy; since consecutive copies are tool-disjoint, entering a new copy rebuilds its first magazine afresh (3 insertions). The cost is 6 insertions for the first copy and 6 for each of the others, i.e. $6g$ in the empty-start count and $6g - 3$ free-initial; hence $Z^* = 6g - 3$.

For the heuristic, a minimum-cardinality grouping of R_g uses $K^* = 3g$ configurations, three per copy. In the ring, those three configurations are pairwise at distance 2, so any ordering of one copy's three contributes $2 + 2 = 4$; and any transition between configurations of different copies costs 3, the copies being tool-disjoint. Connecting g copies needs at least $g-1$ inter-copy transitions, so every ordering costs at least $4g + 3(g-1) = 7g - 3$, attained by keeping each copy's configurations consecutive. Thus $H = 7g - 3$ and $H - Z^* = g$. □

Proof of Proposition 10. Exactness. Take an optimal schedule and collapse equal consecutive magazines to a trajectory $D_1, \ldots, D_m$ ($m \leq n$) as in the proof of Theorem 1. Set $y^l_{D_l} = 1$ for $l = 1, \ldots, m$, leave positions $m{+}1, \ldots, n$ empty, and $w^k_t = \max(0, a^k_t - a^{k-1}_t)$. Then (P′) holds ($= 1$ on the prefix, $= 0$ after) and (G) holds (the prefix is contiguous), while (Cov), (W), (T) hold exactly as for PCF; the full-to-empty step $m \to m{+}1$ and all later steps add no cost, so the objective is $\sum_{l<m} d(D_l, D_{l+1}) = Z^*$. Hence the optimum is at most Z^*. Conversely, any integer-feasible PCF′ point has, by (G), its used positions in a prefix $1, \ldots, p$, each holding one configuration D_k; by (W) the objective is at least $\sum_{k=2}^{p} d(D_{k-1}, D_k)$, and by (Cov) every job is served by some D_k, so assigning each job to a covering position yields a schedule of cost at most the objective. Hence the optimum is at least Z^*.

LP invariance. Every PCF-feasible point has $\sum_C y^k_C = 1$ for all k and so satisfies (P′) and (G); the PCF′ relaxation thus minimises over a superset and is no larger than PCF's. For the reverse, summing (T) over $t \in U$ and using $\sum_{t \in U} a^1_t \leq \sum_{t \in T} a^1_t = b \sum_C y^1_C \leq b$ (the last bound now from (P′) rather than (P)) gives objective $\geq |U| - b$ at every PCF′ point, so the two relaxations coincide wherever PCF's value is $|U| - b$. The plain relaxation, dropping (T), is certified 0 by the same blended configuration used for PCF. □

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